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1. Explain the relationship between binary notation and the physical components of a computer, such as transistors. Why is binary notation the fundamental language of computers, and how does this concept apply to the digital representation of all types of data, including text, images, and sound?
 - A binary system uses 0s and 1s to represent all information, and these values correspond to the off and on positions of transistors. Thus, binary code is the basic language of the computer. The text, pictures, and sounds must be converted to binary code so that a computer can store or manipulate them.
2. Our lesson describes hexadecimal notation as a more efficient way to represent large values than binary. Describe a real-world scenario in which hexadecimal notation is used, and explain why it is more practical for humans to use in that context.
 - Hexadecimal is used by programmers that are dealing with computer values like the address in memory. It is more convenient to use hexadecimal because it is a shorter representation of binary numbers.
3. Our lesson discusses several forms of intellectual property, including copyright, trademark, and patent. Choose one of these forms of protection and explain a time when you have encountered it in your daily life, such as in a software application, a logo, or a product design.
 - I encounter copyright when using software applications. The original software and its content are protected, so other people cannot simply copy or distribute them as their own.
4. Our lesson states that data is used to drive strategic business decisions in areas like sales and marketing. Provide a specific example of how a company could use data analytics to improve its marketing campaign for a product you are familiar with.
 - A company selling smartphones could analyze customer data to see which features are most popular. If customers are mostly interested in the camera, the company could focus its advertisements on the camera features.

5. Explain the difference between a bit, a byte, a kilobyte, and a gigabyte, and provide a real-world example of a file that would be measured in each of these units. Why is it important to understand these units when dealing with a computer's storage capacity?
 - A bit is one binary digit; a byte is eight bits. A kilobyte (KB) is 1,000 bytes. A gigabyte (GB) is 1,000,000,000 bytes. In case some people have forgotten about the binary system, a megabyte (MB) is 1,000,000 bytes. For example, a text file could take up space on a computer's memory in KB's while a video might take up space in GB. This information is helpful because we now know how big various files take up in memory.
6. Our lesson describes various types of software licenses, including proprietary, freeware, shareware, and open source. Choose two of these license types and explain how a developer's choice of license could impact both their business model and a user's experience with the software.
 - Proprietary software is software that is owned and controlled by its developer. Usually, there are limitations and restrictions on its usage. Open-source software is software whose source code is made public, giving users more freedom and flexibility. These differences have an impact on the ways of earning profits from the two types of software and the ways of using them.
7. Explain the importance of a proper uninstallation process for an application. What are the potential consequences of simply deleting an application's files from the *Program files* folder and how does this relate to the concept of "orphaned" files and registry entries?
 - Proper uninstallation removes the application's files and prevents their accumulation in the system in the future. Uninstalling an application by deleting its files in the Program Files directory may leave registry entries and other files in the system.
8. Our lesson discusses several best practices for installing software, such as checking for OS and hardware compatibility. Describe a hypothetical situation where a user ignores these practices, and explain the specific problems they might encounter as a result.
 - If the user installs any software without checking compatibility, they can expect malfunctioning, bad performance, or even a failure to install. The computer may also not have the right hardware. Checking the operating system and requirements can prevent all of these issues.
9. Our lesson categorizes software into different types, such as productivity, collaboration, and business software. Choose one specific application (e.g., Google Docs, Microsoft Teams) and describe how it can be used for both productivity and collaboration, providing examples of its features in each context.

- Google Docs can be used for productivity by creating and editing school documents. It can also support collaboration because multiple people can work together on shared documents, making it useful for group projects.

10. What is the purpose of an End-User License Agreement (EULA)? Describe a scenario where a user might encounter a EULA, and explain why it is a legally binding contract that defines the terms of use for the software.

- A EULA explains the rules and terms for using software. For example, I may encounter one when installing an application and must accept it before continuing. It defines what users are allowed to do with the software and serves as a legal agreement.